

Hail Song

PHD STUDENT · KAIST UVR LAB

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Education

Korea Advanced Institute of Science and Technology

PH.D. STUDENT, GRADUATE SCHOOL OF CULTURE TECHNOLOGY

• Advisor: Prof. Woontack Woo

Daejeon, Republic of Korea

Mar 2024 - present

Korea Advanced Institute of Science and Technology

M.S., GRADUATE SCHOOL OF CULTURE TECHNOLOGY

• Advisor: Prof. Woontack Woo

• Thesis : Real-time Cumulative SMPL-based Avatar Body Generation System

Daejeon, Republic of Korea

Mar 2022 - Feb 2024

Yonsei University

B.S., MECHANICAL ENGINEERING

Seoul, Republic of Korea

Mar 2015 - Feb 2021

Experiences

- Aug 2024 - **Visiting Scholar at Carnegie Mellon University**, Sponsored by Institute of Information & Communications Technology Planning & Evaluation, IITP.
- Feb 2025 - **Visiting Scholar at TUM (Technical University of Munich)**, TUM School of Computation, Information and Technology, CAMP
- Jul 2023 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- Aug 2023 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- Mar 2022 - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- present - **Research Assitant**, Development of Untact Realistic OpenXR Platform Technology. National Research Project funded by Korean Government. Joint Research Project with KISTI, KAIST, KICT, and KIOM.
- May 2020 - **Computer Vision Researcher (Full-time)**, InBody Co., Ltd.
- Feb 2022 - **Computer Vision Researcher (Full-time)**, InBody Co., Ltd.
- Oct 2016 - **Mandatory Military Service**, ROK Army sergeant, honorable discharge
- Jul 2018 - **Mandatory Military Service**, ROK Army sergeant, honorable discharge

Research Interests

AVATAR CREATION WITH NEURAL RENDERING

I am dedicated to researching efficient 3D avatar generation. My work focuses on creating high-fidelity avatars from minimal user input, as demonstrated in my research on head avatar reconstruction from a single image using Diffusion and 3DGS [Under review], and real-time full-body avatar generation from monocular RGB-D video [ISMAR 2023]. More recently, I have also explored generating 3D face models from 3D sketches [CVPRW 2025]. My broader interests lie in the diverse applications of neural rendering and advancing 3D human model representations. Ultimately, my goal is to enhance the accessibility of this technology, enabling all users to freely and equitably express their identities in the metaverse.

AVATAR INTERACTION ON XR DEVICES

Recent advancements in neural rendering, including NeRF and 3D Gaussian Splatting, enable photo-realistic avatar creation. However, deploying these on consumer HMDs like the Meta Quest or Apple Vision Pro presents significant challenges. Key obstacles include maintaining novel-view consistency, tracking facial expressions with internal sensors, and estimating body pose from egocentric-only data, all within limited computational budgets. To address these issues, my current research focuses on developing an integrated metaverse communication system that delivers seamless and realistic avatar interactions on XR devices.

Publications_____

[Under Review] Diffusion and 3DGS-based Head Avatar Reconstruction

Hail Song*, et al.
Submitted to a top-tier AR/VR conference

S3D: Sketch-Driven 3D Model Generation

Hail Song*, Wonsik Shin*, Naeun Lee, Soomin Chung, Nojun Kwak, Woontack Woo
GMCV Workshop at CVPR'25 (**CVPRW 2025**)

The Influence of Emotion-based Prioritized Facial Expressions on Social Presence in Avatar-mediated Remote Communication

Seoyoung Kang, **Hail Song**, Boram Yoon, Kangsoo Kim, Woontack Woo
2024 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**)

Toward Realistic 3D Avatar Generation with Dynamic 3D Gaussian Splatting for AR/VR Communication

Hail Song
2024 IEEE Virtual Reality Workshop (Doctoral Consortium) (**VRW 2024**)
Best DC Paper Awards

Meta-Rodin: a Multi-Sensory Immersive Sculpting System to Cultivate a Memorable Experience in a Meta-Museum

Sihyun Jeong, Coulon Jina Jung, **Hail Song**, Woontack Woo
Korea Computer Congress 2024 (**KCC 2024**)
Outstanding Presented Paper Award

RC-SMPL : Real-time Cumulative SMPL-based Avatar Body Generation

Hail Song, Boram Yoon, Woojin Cho, Woontack Woo
2023 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2023**)

Quantitative Optimal Learning Conditions for Gaussian Splatting Based On Input Quantity and Scene Dynamics

Hyungwoo Jin, Eunhee Jung, **Hail Song**, Woontack Woo
Korea Software Congress 2023 (**KSC 2023**)

Effects of Different Facial Blendshape Combinations on Social Presence for Avatar-mediated Mixed Reality Remote Communication

Seoyoung Kang, **Hail Song**, Boram Yoon, Kangsoo Kim, Woontack Woo
2023 IEEE International Symposium on Mixed and Augmented Reality (**ISMAR-Adjunct 2023**)

Real-Time Music Tracking System in Wearable AR Display for Music Performance

Jiyun Park, **Hail Song**, Seungmin Lee, Juhan Nam, Woontack Woo
2023 The HCI Society of Korea (**HCI Korea 2023**)

Awards, Fellowships, & Grants_____

2024	IEEE VR 2024 Best DC Paper Award, IEEE VR 2024	
2024	KCC Outstanding Presented Paper Award, KCC 2024	
2020	Department Honors, Yonsei University	
2015	National Science & Technology Scholarship, KOSAF (Korea Student Aid Foundation)	\$ 26,000

Teaching Experience_____

2025	CTP445 Augmented Reality, Teaching Assistant	KAIST, Daejeon
2019	Teaching Assistant (Coding), Samsung Dream Class, supported by Samsung Co., Ltd.,	Incheon

Academic Activities_____

INVITED TALKS

2023 Korea Software Congress (KSC), Top-Conference Session

REVIEWER

IEEE ISMAR 2024, 2025 / Korea Computer Congress (KCC) 2024, 2025